

Adaptive Multi-Interaction Web Semantic Graph Representation

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Abstract

Effective representations of complex web semantic graphs are essential for various web applications, including link prediction, recommendation systems, and social network analysis. However, existing methods assume that multi-interactions (or multi-relationships) between two connected nodes are independent, while these relationships inherently exhibit characteristics of *mutual promotion* or *mutual inhibition*. Moreover, these semantic characteristics across different relationships cannot be easily captured by a simple linear combination. To tackle this challenge, we propose an Adaptive Multi-Interaction (AMI) web semantic graph representation method. Specifically, AMI consists of three modules, including a multi-interaction aggregation module, a global pattern aggregation module, and an adaptive relation-specific decoder module. Firstly, we construct a learnable multi-interaction behavior pattern matrix that captures the mutual promotion and mutual inhibition effects between two connected nodes. Secondly, the global pattern aggregation module is designed to efficiently capture global homogeneous interaction patterns through graph convolution networks. Finally, the adaptive relation-specific decoder module employs a hybrid scoring strategy to adaptively decode node embeddings based on their distinct relationships. Extensive experiments on benchmark web datasets for link prediction tasks demonstrate that AMI outperforms state-of-the-art baselines. Our codes are available at <https://github.com/AI-stronger123/AMI>.

CCS Concepts

• **Computing methodologies** → **Semantic networks.**

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Keywords

Graph representation learning, heterogeneous networks, knowledge fusion, graph neural networks

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1 Introduction

In real-world scenarios, web semantic graphs are often considered multi-relational heterogeneous networks, consisting not only of multi-typed nodes but also of diverse relationships between the same pairs of nodes [8, 41]. In E-commerce networks, there exist multi-interactions (i.e., click, favorite, add-to-cart, and purchase) between the same pairs (i.e., users and items) [38]. Compared to traditional heterogeneous networks, multi-relational heterogeneous networks are more complex. The reason is that various relationships reveal distinct semantic information and it is hard to model interactions of target nodes in web networks [22, 28]. To effectively capture their various semantic knowledge, multiplex heterogeneous network embedding methods have been proposed, which have demonstrated great success in link prediction, recommendation systems, and social network analysis [13, 23, 37, 40].

Existing methods aim to learn low-dimensional node embeddings by leveraging structural and semantic information from different relationships between two connected nodes [25, 28]. Inspired by the effectiveness of Graph Neural Networks (GNNs) in aggregating local information from neighboring nodes, numbers of GNN-based heterogeneous graph algorithms have been proposed [5]. For example, some approaches convert heterogeneous graphs into homogeneous ones based on different meta-paths and aggregate neighboring information through GNNs [3, 16, 18]. However, such approaches often overlook the complex and diverse relationships between nodes and edges, leading to the loss of critical information

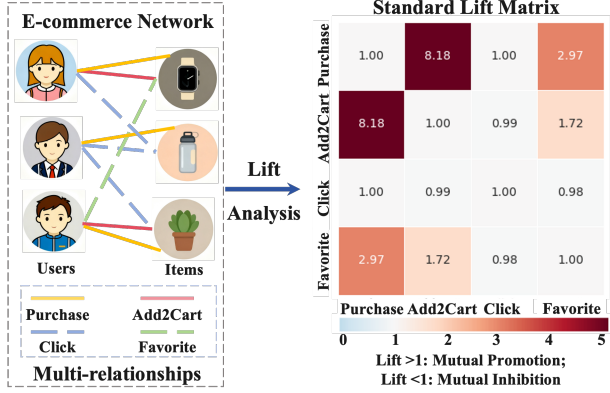


Figure 1: An illustrative example highlighting the mutual promotion or mutual inhibition characteristics in an E-commerce network.

[41]. To solve this problem, Shen et al. [27] proposed an information-aware unsupervised multiplex graph fusion framework that learns a fuse graph from the original multiplex graph to reduce task-irrelevant noise.

Although these studies have achieved great success, they neglect that the simultaneous multi-relationships among any pairs of nodes exhibit *mutual promotion* or *mutual inhibition* effects. To quantify these two characteristics of multiple relationships, we employ the standard lift analysis in this paper [1, 30]. As shown in Figure 1, associations between four types of relationships are quantified via a standard lift matrix in the Alibaba-s dataset [19], a real-world web semantic graph. For example, behavior *purchase* and behavior *add2cart* are mutually promoted, indicating that a user's *add2cart* behavior will highly encourage him to purchase the item [15]. In contrast, the *click* behavior is mutually inhibited by the *favorite* behavior, indicating that the occurrence of a *click* reduces the likelihood of a *favorite* being made. Apparently, various pairs of interactions show different mutual promotion or inhibition characteristics and should not be treated independently. If treating multi-relationships independently, it will lead node representations hard to preserve the distinct structural and semantic information from various multi-interaction views. However, it is a challenge to capture mutual promotion and mutual inhibition characteristics of multi-relationships among various pairs of nodes.

In this paper, we propose an **Adaptive Multi-Interaction** representation (**AMI**) method to tackle the above challenge. Specifically, AMI consists of three modules, including a multi-interaction aggregation module, a global pattern aggregation module, and an adaptive relation-specific decoder module. In the first module, AMI effectively captures mutual promotion or mutual inhibition effects of multiple relationships based on a learnable multi-interaction behavior pattern matrix. To capture global information well, AMI fuses homogeneous interaction patterns through multilayer Graph Convolution Networks (GCNs) in the second module. Finally, we use a hybrid scoring strategy that combines general similarity and specific relation scores to capture distinct semantic information of multiplex relationships between various node pairs. To evaluate the

effectiveness of our AMI, we conduct experiments on five public web semantic graph datasets. Experimental results demonstrate that our proposed AMI outperforms state-of-the-art baselines in link prediction.

We summarize the contributions of this paper as follows:

- We propose an adaptive multi-interaction web semantic graph representation method, namely AMI, which effectively captures the mutual promotion or mutual inhibition characteristics in the multi-relational heterogeneous network.
- We construct a learnable multi-interaction behavior pattern matrix to explicitly model mutual promotion or promotion inhibition characteristics of multi-interactions.
- We conduct extensive experiments on several real-world web datasets. The experimental results demonstrate that our proposed AMI outperforms the state-of-the-art baselines in link prediction tasks.

2 Related Work

Heterogeneous Network Embedding methods can effectively learn underlying semantic information of various types of nodes and edges [6, 29, 31]. For example, Fu et al. [11] proposed a method named MECCH, which leverages meta-path contexts to capture structural and semantic information for efficient information fusion. To further reduce the information loss caused by ignoring intermediate nodes, Fu et al. [12] proposed to employ an intra-path aggregation mechanism that encodes intermediate nodes with the integration of rich semantic information. Unlike the previous approaches, message passing-based approaches directly propagate information among heterogeneous neighbors rather than predefined meta-paths. For example, Michael et al. [26] proposed a method named R-GCN, which effectively captures various relationships through a relation-specific transformation matrix for each type of edge. Xue et al. [32] and Zhang et al. [21] proposed using attention mechanisms to capture information from one-hop and high-order neighbors. However, many real-world networks consist not only of multi-typed nodes and edges but also of diverse edges between the same pair-wise nodes. To capture these heterogeneity properties, multiplex heterogeneous network embedding studies have received some attention.

Multiplex Heterogeneous Network Embedding methods are primarily divided into two categories, i.e., relation decomposition with independent aggregation approaches and meta-path aggregation approaches. The former approaches aim to partition a multiplex graph into single-relation subgraphs and fuse multiple subgraph information together. For example, Yu et al. [36] proposed to transform the multi-relational based on various relation types. Cen et al. [2] proposed to project networks into a shared embedding space and employ a self-attention mechanism to aggregate them into a unified representation. Park et al. [24] introduced a consensus regularization framework to integrate embeddings, which maximizes the mutual information between local and global views for each relationship. Xue et al. [33] proposed to transform bi-partite networks into hyper-graphs and capture complex relationships through hypergraph convolution networks. Meta-path aggregation approaches aim to capture higher-order semantic information by integrating relations through diverse paths [14]. For example, Fu et al.

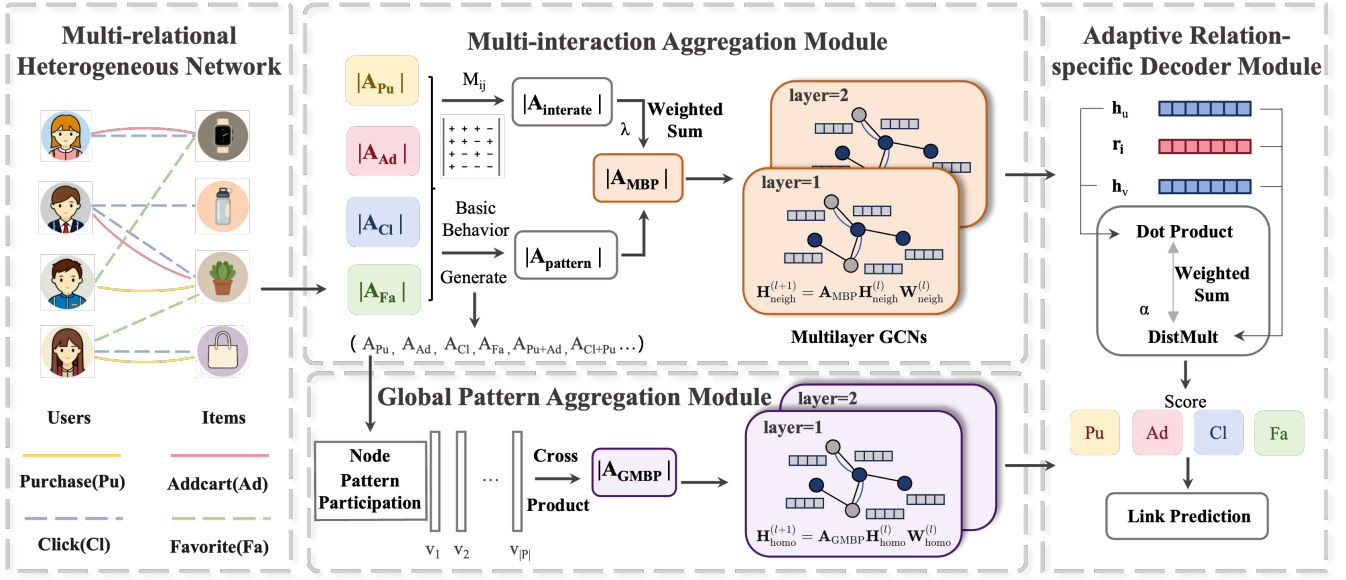


Figure 2: Overall architecture of our AMI.

[10] introduced basic behavior patterns by combining all multiple relationships between two connected nodes into unified representations with different weights. Liu et al. [20] designed relation-aware meta-paths that can be applied to large networks through spectral graph transformation. Fu et al. [9] proposed to model high-order interactions in multiplex heterogeneous networks to solve the weight conflict problem in decomposition-based models.

Although existing methods have achieved great success, multi-relationships actually have mutual promotion or mutual inhibition characteristics. To capture these characteristics, we propose an adaptive multi-interaction semantic graph representation method.

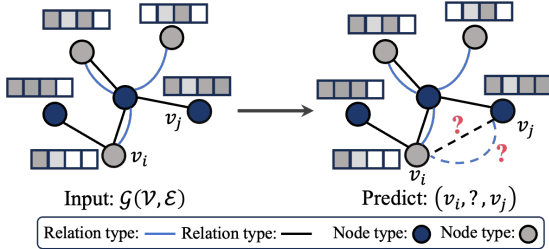


Figure 3: An illustration of link prediction task.

3 Preliminaries

Definition: Multi-relational Heterogeneous Network. We consider a network as a multi-relational heterogeneous network defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ represents the set of nodes and $\mathcal{E}$ denotes the set of edges. The node set $\mathcal{V}$ and the edge set $\mathcal{E}$ are assigned different types through the mapping functions $\phi: \mathcal{V} \rightarrow N$ and $\psi: \mathcal{E} \rightarrow R$, where $|N| + |R| > 2$. In addition, multiple types of edge are allowed to exist between the same pair of nodes. For any two nodes $v_i, v_j \in \mathcal{V}$, there exists a set of edges $\{e_1, e_2, \dots, e_k\} \subseteq \mathcal{E}$

where $k \geq 2$, and each edge e_t between the pair of nodes (v_i, v_j) represents a different type of relationship.

Task Description: Link Prediction. In this work, the link prediction task is used to verify the effectiveness of multi-relational heterogeneous network embeddings. Link prediction aims to predict missing links between two nodes by leveraging their topology structures and attributes in the network. An example of link prediction between node v_i and node v_j is shown in Figure 3, the object is to infer missing connections, i.e., (v_i, r_1, v_j) and (v_i, r_2, v_j) in the network $\mathcal{G}$, where r_1 and r_2 denotes two different types of relations.

Owing to page limits, the key notions used in this paper and the definition of lift value are listed in the appendix A.1.

4 Methodology

In this section, we present the details of our proposed AMI method. As illustrated in Figure 2, the overall architecture of AMI is the encoder-decoder architecture. Specifically, AMI contains three modules: (i) a *multi-interaction aggregation module*, (ii) a *global pattern aggregation module*, and (iii) an *adaptive relation-specific decoder module*. The first two modules serve as the encoder to learn node embeddings by capturing multi-interaction characteristics in multi-relational heterogeneous networks. Firstly, the *multi-interaction aggregation module* introduces a learnable multi-interaction behavior pattern matrix, which can efficiently capture the mutual promotion and mutual inhibition effects of multiple relations. After obtaining the multi-interaction behavior pattern matrix, the *global pattern aggregation module* aggregates information based on the same behavior patterns through multilayer GCNs, which serves as a complement to well integrate local and global multi-interactions of target nodes. Finally, the *adaptive relation-specific decoder module* is designed by a hybrid scoring strategy that combines general similarity and relation-specific scores to provide more accurate predictions for different types of relationships.

4.1 Multi-interaction Aggregation

To capture the mutual promotion and mutual inhibition characteristics of multi-relationships, AMI models multi-interactions between various node pairs in multi-relational heterogeneous networks. Inspired by BPHGNN [10], AMI first extracts the basic behavior patterns between two connected nodes in the network. The basic behavior pattern consists of a series of basic behaviors that are extracted from various types of relationships in the networks. For example, in the E-commerce network, *Purchase*, *Addcart*, *Click* and *Favourite* are the basic behaviors. If a user first clicks on an item and finally purchases it, the logical variables of the basic behavior pattern would be denoted as $\{click : 1, Addcart : 0, Purchase : 1, Favourite : 0\}$.

According to various types of relationships in multi-relational heterogeneous networks, they can be decomposed into various homogeneous networks. And they can be described as a set of basic behavior matrices $\mathbf{A}_{i=1,2,\dots,|R|} \in \mathbb{R}^{|V| \times |V|}$, where $|V|$ denotes the number of all nodes and $|R|$ denotes the number of all relation types in the network. The basic behavior pattern matrices $\mathbf{A}_{\text{pattern}} \in \mathbb{R}^{|V| \times |V|}$ can be generated through basic behavior matrices and weight operations in parallel, which can be denoted as Equation 1.

$$\mathbf{A}_{\text{pattern}} = \sum_{i=1}^N \alpha_i \cdot \mathbf{A}_i, \quad (1)$$

where α_i is a learnable weight parameter, N is the number of basic behavior patterns, and $\mathbf{A}_i$ denotes the basic behavior matrix.

Next, we construct a learnable interaction-enhanced matrix to capture the strength of mutual promotion or mutual inhibition effects between various pairs of relationships. Taken basic behavior matrix $\mathbf{A}_i$ and $\mathbf{A}_j$ as an example, the interaction-enhanced matrix $\mathbf{I}_{\text{mult}}^{(i,j)} \in \mathbb{R}^{|V| \times |V|}$ is formulated as Equation 2.

$$\mathbf{I}_{\text{mult}}^{(i,j)} = \mathbf{W}_1 \cdot \mathbf{A}_i \odot \mathbf{A}_j, \quad (2)$$

where $\mathbf{W}_1 \in \mathbb{R}^{|V| \times |V|}$ is a learnable parameter matrix and $\odot$ denotes the Hadamard product [4].

After obtaining the strength of multiplex relationships, we construct the conditional interaction matrix to determine the direction of these mutual effects. If the existence of relation r_i promotes the occurrence of relation r_j , the direction is positive. Otherwise, the direction is negative. Taken the relation r_i and relation r_j as an example, the conditional interaction matrix $\mathbf{I}_{\text{cond}}^{(i,j)} \in \mathbb{R}^{|V| \times |V|}$ is formulated as Equation 3.

$$\mathbf{I}_{\text{cond}}^{(i,j)} = \mathbf{W}_2 \cdot \mathbb{I}(\mathbf{A}_i) \odot \mathbf{A}_j, \quad (3)$$

where $\mathbf{W}_2 \in \mathbb{R}^{|V| \times |V|}$ is a learnable parameter matrix and $\mathbb{I}(\cdot)$ is an indicator function.

To prevent the items in the matrices from being extremely large, we used an active function to normalize them. And the enhanced interaction adjacency matrix $\mathbf{A}_{\text{interact}} \in \mathbb{R}^{|V| \times |V|}$ is formulated as Equation 4.

$$\mathbf{A}_{\text{interact}} = \sigma \left(\sum_{i=1}^{|V|} \sum_{j=1, j \neq i}^{|V|} [w_1 \cdot \mathbf{I}_{\text{mult}}^{(i,j)} + w_2 \cdot \mathbf{I}_{\text{cond}}^{(i,j)}] \right), \quad (4)$$

where $\sigma(\cdot)$ is an active function that bounds the values within $[-1, 1]$ and $w_1, w_2 \in (0, 1)$ are hyperparameters that balance the importance between two matrices.

The multi-interaction behavior pattern matrix $\mathbf{A}_{\text{MBP}} \in \mathbb{R}^{|V| \times |V|}$ is formulated as Equation 5.

$$\mathbf{A}_{\text{MBP}} = \mathbf{A}_{\text{pattern}} + \lambda \cdot \mathbf{A}_{\text{interact}}, \quad (5)$$

where $\mathbf{A}_{\text{pattern}} \in \mathbb{R}^{|V| \times |V|}$ is the behavior pattern matrices, λ is a learnable parameter that controls the importance of two matrices, and $\mathbf{A}_{\text{interact}} \in \mathbb{R}^{|V| \times |V|}$ is the enhanced interaction adjacency matrix.

Then, we employ multilayer GCNs to aggregate information of multiple interactions in the multi-relational heterogeneous network as Equation 6.

$$\mathbf{H}_{\text{neigh}}^{(l+1)} = \mathbf{A}_{\text{MBP}} \mathbf{H}_{\text{neigh}}^{(l)} \mathbf{W}_3^{(l)}, \quad (6)$$

where $\mathbf{W}_3^{(l)} \in \mathbb{R}^{n_2 \times |V|}$ is a learnable parameter matrix of the l -th layer. The initial node representation $\mathbf{H}_{\text{neigh}}^{(0)} \in \mathbb{R}^{|V| \times n_2}$ is the original node feature matrix $\mathbf{X} \in \mathbb{R}^{|V| \times n_1}$.

Finally, we obtain the node embeddings $\mathbf{H}_{\text{neigh}} \in \mathbb{R}^{|V| \times n_2}$, aggregating multi-interaction information from the local perspective, which can be described as Equation 7.

$$\mathbf{H}_{\text{neigh}} = \frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\text{neigh}}^{(l)}, \quad (7)$$

where L is the total number of GCN layers and n_2 is the dimension of node embeddings in this module.

4.2 Global Pattern Aggregation

To capture global structural information in the multi-relational heterogeneous network, AMI aggregates information through various basic behavior patterns in parallel. For example, the target node, such as users who have the "click-and-purchase" behavior pattern, influences the representation of multi-hop nodes even if they are not directly connected.

Specifically, given a node v_i , its global embedding vector $\mathbf{v}_{\text{pattern}(i)} \in \mathbb{R}^{|V| \times 1}$ indicates that it aggregates information from all the other nodes $v_{j=1,2,\dots,|V|}$ that have the same basic behavior pattern, and the vector $\mathbf{v}_{\text{pattern}(i)}$ can be formulated as Equation 8.

$$\mathbf{v}_{\text{pattern}(i)} = \sum_{j=1}^{|V|} \mathbf{A}_{\text{pattern}(i,j)}. \quad (8)$$

Then, we use a set of learnable weights β_i to concatenate all global embedding vectors based on various patterns to obtain the node-level behavior matrix $\mathbf{A}_n \in \mathbb{R}^{|V| \times |P|}$ as Equation 9.

$$\mathbf{A}_n = [\beta_1 \cdot \mathbf{v}_1 \parallel \beta_2 \cdot \mathbf{v}_2 \parallel \dots \parallel \beta_{|P|} \cdot \mathbf{v}_{|P|}], \quad (9)$$

where $|P|$ is the total number of basic behavior patterns, and $[\cdot \parallel \cdot]$ denotes the concatenation operation.

Next, we multiply the node-level behavior matrix by its transpose and normalize it to obtain the global behavior pattern matrix $\mathbf{A}_{\text{GBPM}} \in \mathbb{R}^{|V| \times |V|}$ as Equation 10.

$$\mathbf{A}_{\text{GBPM}} = \text{Normalize}(\mathbf{A}_n \cdot \mathbf{A}_n^T). \quad (10)$$

Similar to the former module, we input the global behavior pattern matrix into multilayer GCNs to obtain global node representations

$\mathbf{H}_{homo} \in \mathbb{R}^{|V| \times n_2}$. To well obtain representations based on local and global information, we use a double information-fused mechanism that leverages local multi-interaction patterns and global behavior patterns.

Finally, node embeddings $\mathbf{H}_{final} \in \mathbb{R}^{|V| \times n_2}$ can be formulated as Equation 11.

$$\mathbf{H}_{final} = \delta \cdot \mathbf{H}_{neigh} + (1 - \delta) \cdot \mathbf{H}_{homo}, \quad (11)$$

where $\mathbf{H}_{neigh} \in \mathbb{R}^{|V| \times n_2}$ are learnable embeddings that capture both mutual promotion and mutual inhibition effects, $\mathbf{H}_{homo} \in \mathbb{R}^{|V| \times n_2}$ is an embedding matrix based on the global behavior patterns, and $\delta \in (0, 1)$ is a hyperparameter that balances the importance of local and global information. In this paper, we set $\delta = \frac{1}{2}$. To stabilize the embedding space, we perform L2 normalization on the node embeddings $\mathbf{H}_{final}$, and the final node embeddings are denoted as $\hat{\mathbf{H}}_{final} = \|\mathbf{H}_{final}\|_{l2}$.

4.3 Adaptive Relation-specific Decoder

To capture multi-interaction patterns across multiple types of relationships in the multi-relational heterogeneous network, we design an adaptive relation-specific decoder. As a knowledge graph embedding method, DisMult [34] shows better performance for multiple types of edges in networks. Thus, we design a hybrid scoring strategy that uses both a general similarity-based component and a relation-specific component. The former component captures the general similarity between nodes in the embedding space, while the latter introduces relation-specific inductive biases. Combining these components can improve the accuracy of link prediction in the multi-relational heterogeneous network.

Firstly, we employ the conventional dot product to measure the general similarity between the source node v_s and target node v_t . The general similarity score $s_{dot}(v_s, v_t)$ is formulated as Equation 12.

$$s_{dot}(v_s, v_t) = \mathbf{h}_{v_s}^\top \mathbf{h}_{v_t}, \quad (12)$$

where $\mathbf{h}_{v_s}, \mathbf{h}_{v_t} \in \mathbb{R}^{n_2 \times 1}$ are the embeddings of node pair v_s, v_t , respectively. The general similarity score reflects the basic residual similarity of nodes in the low-dimensional latent embedding space.

Then, we use DistMult [34] to compute the relationship-specific score $s_{dist}(\cdot)$, i.e., for node pair v_s, v_t and a target edge type r_i as Equation 13.

$$s_{dist}(v_s, r_i, v_t) = \text{sigmoid}(\mathbf{h}_{v_s}^\top \mathbf{r}_i \mathbf{h}_{v_t}), \quad (13)$$

where $\mathbf{r}_i \in \mathbb{R}^{n_2 \times 1}$ is embedding of relation r_i through random initialized.

Finally, the hybrid scores of link prediction for node pair v_s, v_t and a target relation r_i is described as Equation 14.

$$s_{final}(v_s, r_i, v_t) = (1 - \alpha) \cdot s_{dot}(v_s, v_t) + \alpha \cdot s_{dist}(v_s, r_i, v_t), \quad (14)$$

where $\alpha \in [0, 1]$ is a hyperparameter that balances the contribution between the general similarity and relationship-specific score.

4.4 Model Training and Regularization

To ensure that the AMI can effectively learn representations in the multi-relational heterogeneous network, we design a comprehensive end-to-end learning objective and incorporate two regularization strategies to guarantee its stability and generalization. The pseudo code of AMI is shown in Algorithm 1 of the appendix.

The link prediction task is formulated as a supervised binary classification problem. For a given pair of nodes, AMI aims to infer whether the target relation exists between various node pairs in the network. In addition, the probability of different types of relations varies significantly. For example, the relation *click* may occur more frequently than another relation *purchase*. If we use a unified loss function across all relations, the model will be biased toward relation types that occur more frequently. In addition, the model may overlook infrequent but important relationships. To solve this challenge, we propose a relation-specific loss function that ensures balanced optimization across all types of relationships.

Given an example of relation r_i , its binary cross-entropy loss function $\mathcal{L}_{r_i}$ is denoted as Equation 15.

$$\mathcal{L}_{r_i} = -\frac{1}{|\mathcal{G}_{r_i}|} \sum_{(v_s, v_t) \in \mathcal{G}_{r_i}} \left[y \log(s_{final}(v_s, r_i, v_t)) + (1 - y) \log(1 - s_{final}(v_s, r_i, v_t)) \right], \quad (15)$$

where $\mathcal{G}_{r_i}$ is the sample of $\mathcal{G}$ for the relation r_i , $y \in \{0, 1\}$ is the ground-truth label of the edge in $\mathcal{G}_{r_i}$. If the edge between the node pair (v_s, v_t) exists, the label $y = 1$; otherwise, $y = 0$. $s_{final}(\cdot)$ denotes the probability of predicting the existence of the edge.

Then, the final link prediction loss $\mathcal{L}_{link}$ of all types of relations is obtained through an average operation as Equation 16.

$$\mathcal{L}_{link} = \frac{1}{|R|} \sum_{r_i=1}^{|R|} \mathcal{L}_{r_i}, \quad (16)$$

where $|R|$ is the total number of all relation types. To prevent the model from being overfit, we add an L_2 regularization [7] to the overall loss. The final loss function $\mathcal{L}_{total}$ is formulated as Equation 17.

$$\mathcal{L}_{total} = \mathcal{L}_{link} + \gamma \|\theta\|_2^2, \quad (17)$$

where γ is the hyperparameter to balance these two items in the loss function.

4.5 Complexity Analysis

We analyze the complexity of AMI by taking three main components into consideration respectively. For the first module, the time complexity is $O(m^2|V|^2)$, where m denotes the total number of relationships and $|V|$ represents the total number of nodes. In addition, the multi-interaction aggregation module includes the construction of the pattern aggregation matrix $A_{pattern}$ and GCN information aggregation for node learning. Hence, the time complexity of the $A_{pattern}$ construction is $O(|V|^2|P|)$, where $|P|$ denotes the total number of basic behavior patterns. The time complexity of GCN information aggregation is $O(|V||P|)$. For the global pattern aggregation module, the whole time complexity is $O(|V|^2|P| + |V||P|)$. The time complexity of the decoder module is $O(|V|^2|\mathcal{E}||R|)$, where $\mathcal{E}$ denotes the total number of edges and R represents the types of relations. All in all, the total complexity of AMI is $O(m^2|V|^2 + 2|V|^2|P| + 2|V||P| + |V|^2|\mathcal{E}||R|)$.

5 Experiments

In this section, we first present comparison experiments of AMI on the link prediction tasks. Then, we extend our study with ablation experiments and the parameter analysis. We adopt two evaluation

Table 1: Link prediction performance comparisons of different baselines on five datasets.

Model	Alibaba-s		Alibaba		Douban		DBLP		IMDB	
	AUC	PR	AUC	PR	AUC	PR	AUC	PR	AUC	PR
HAT	0.594±0.0011	0.582±0.009	0.587±0.017	0.579±0.008	0.601±0.013	0.596±0.010	0.651±0.019	0.562±0.021	0.703±0.005	0.698±0.015
R-GCN	0.683±0.031	0.708±0.022	0.702±0.017	0.698±0.024	0.652±0.033	0.684±0.028	0.632±0.037	0.578±0.025	0.815±0.015	0.853±0.009
SR-RSC	0.866±0.014	0.842±0.010	0.839±0.021	0.823±0.017	0.761±0.027	0.775±0.018	0.748±0.030	0.724±0.026	0.869±0.011	0.871±0.008
MAGNN	0.839±0.008	0.834±0.007	0.864±0.006	0.838±0.007	<u>0.794±0.007</u>	<u>0.815±0.004</u>	0.649±0.006	0.665±0.008	0.833±0.002	0.814±0.003
Simple-HGN	0.897±0.003	0.834±0.011	0.851±0.007	0.830±0.014	0.761±0.008	0.771±0.019	0.761±0.017	0.718±0.012	0.893±0.010	0.927±0.006
MECCH	0.820±0.018	0.831±0.021	0.807±0.024	0.817±0.019	0.775±0.027	0.763±0.020	0.764±0.016	0.681±0.019	0.909±0.012	0.931±0.013
GATNE	0.797±0.018	0.823±0.016	0.737±0.012	0.762±0.008	OOM	OOM	0.721±0.010	0.639±0.014	0.843±0.020	0.802±0.010
FAME	0.692±0.003	0.685±0.005	0.636±0.007	0.674±0.004	<u>0.788±0.003</u>	0.760±0.004	0.626±0.006	0.593±0.112	<u>0.937±0.005</u>	<u>0.956±0.008</u>
DMGI	0.859±0.007	0.845±0.010	0.858±0.005	0.779±0.005	OOM	OOM	0.632±0.011	0.647±0.016	0.927±0.004	0.938±0.003
DualHGCN	0.881±0.005	0.838±0.009	0.852±0.005	0.800±0.004	0.747±0.011	0.775±0.009	\	\	\	\
MHGCN	0.833±0.037	<u>0.874±0.021</u>	<u>0.868±0.009</u>	0.842±0.003	0.734±0.081	0.744±0.013	0.743±0.020	<u>0.703±0.003</u>	0.728±0.003	0.787±0.002
BPHGNN	0.925±0.009	0.869±0.015	0.795±0.003	0.782±0.004	0.705±0.012	0.758±0.003	<u>0.842±0.014</u>	0.689±0.019	0.921±0.022	0.914±0.003
MHGCN-plus	0.681±0.015	0.660±0.020	0.839±0.010	0.839±0.010	0.776±0.008	<u>0.802±0.041</u>	0.713±0.009	0.697±0.008	0.872±0.009	0.841±0.005
DCMGNN	<u>0.928±0.002</u>	0.853±0.012	0.789±0.006	0.778±0.005	0.685±0.014	0.709±0.009	0.736±0.010	0.619±0.012	0.787±0.026	0.707±0.005
AIMS	0.937±0.005	0.875±0.022	0.869±0.005	<u>0.840±0.004</u>	0.797±0.003	0.820±0.005	0.894±0.009	0.729±0.007	0.961±0.001	0.958±0.002

metrics: Area Under the ROC Curve (AUC) and Precision-Recall (PR), which are widely used to evaluate the effectiveness of link prediction tasks. AUC is used to measure the ability to discriminate between positive and negative samples, and PR measures the proportion of false predictions in imbalanced positive sample datasets.

5.1 Experimental Settings

5.1.1 Datasets. To fairly evaluate the performance of the proposed AMI, we use five real-world web datasets, including IMDB¹, Alibaba-s², Alibaba³, DBLP⁴, and Douban⁵. Alibaba, Alibaba-s and Douban are multi-relational heterogeneous networks and the other datasets are traditional heterogeneous networks.

5.1.2 Baselines. To demonstrate the effectiveness of AMI, we compare it with several state-of-the-art baselines, which can be divided into heterogeneous network embedding methods and multiplex heterogeneous network embedding methods. The former category includes the following state-of-the-art methods: HAT [32], R-GCN [26], SR-RSC [39], MAGNN [12], Simple-HGN [21] and MECCH [11]. The later includes the following SOTA methods: GATNE [2], FAME [20], DMGI [24], DualHGCN [33], MHGCN [36], BPHGNN [10], MHGCN-plus [9] and DCMGNN [17].

5.1.3 Experimental Setting. Implementations of all benchmark baselines are based on their official codes to ensure the fairness of our experiments. The hardware platform for our experiments is on dual Intel Xeon Gold 5115 (2.40GHz) CPUs, 256GB of RAM, and a dual NVIDIA Tesla V100 (16GB) setup. All experiments are conducted on

Ubuntu 20.04.5 LTS with Python 3.9.21, PyTorch 1.9.1, and CUDA 11.1. We repeat five times for each experiment with different random seeds, and the averaged results are used as the final results in this paper.

5.2 Link Prediction

To evaluate the performance of our model, we compare AMI with several state-of-the-art baselines on the link prediction task. The results are reported in Table 1, where the best performance is highlighted in bold, the second-best performance is underlined, and the OOM refers to out of memory. The first six baselines are heterogeneous network embedding methods, while the remaining baselines are designed for multi-relational heterogeneous networks.

Overall, AMI achieves superior performance across most datasets, demonstrating significant superiority over existing methods on both the AUC and PR metrics. Compared to the heterogeneous network embedding methods, the multiplex heterogeneous network embedding methods always perform better than them. While MHGCN slightly outperforms AMI on the PR metric in the Alibaba, AMI still achieves the most highest performance in the rest datasets. The slash indicates that DualHGCN is only applicable to multiplex bipartite networks that have no interactions between the same type of nodes. Compared to the traditional heterogeneous graph methods, such as MAGNN, Simple-HGN, and MECCH, which rely on meta-path aggregation or a uniform neighbor weight, AMI effectively captures fine-grained relational interactions and learns more discriminative and robust node representations for web semantic graph. Compared with the state-of-the-art baselines, AMI can effectively alleviate potential semantic conflicts of multiple relations and enhance adaptability to complex relational patterns. Overall, these results comprehensively validate the superiority of AMI in modeling various interactions in multi-relational heterogeneous networks. Our model not only achieves a significant boost

¹<https://github.com/RuixZh/SR-RSC>

²<https://tianchi.aliyun.com/competition/entrance/231719/information/>

³<https://github.com/xuehansheng/DualHGCN>

⁴<https://www.dropbox.com/s/yh4grpeks87ugr2>

⁵<https://github.com/7thsword/MFPR-Datasets>

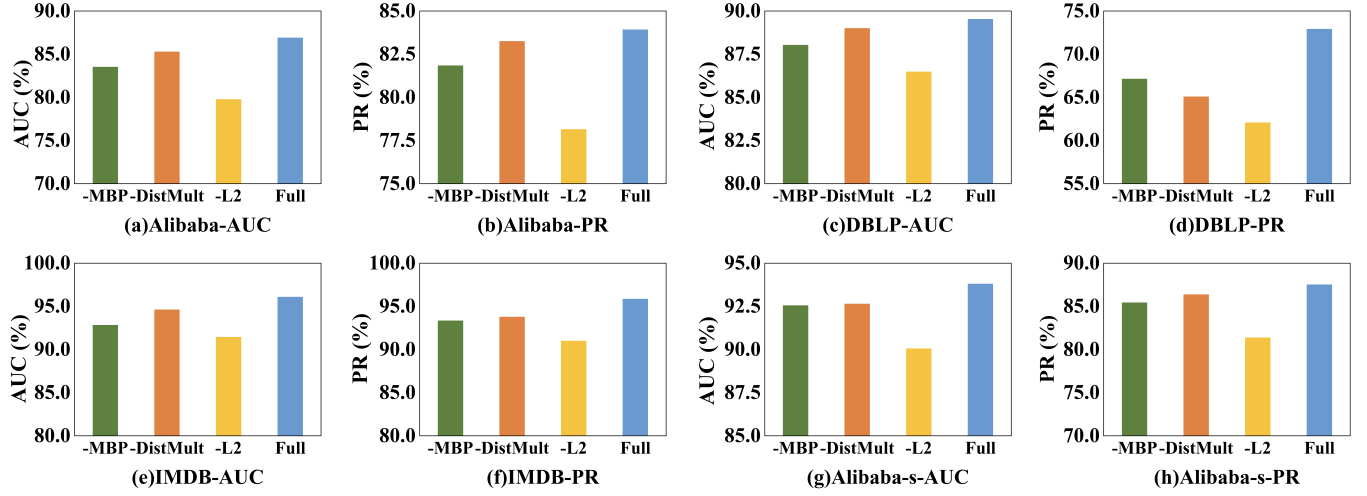


Figure 4: Ablation analysis of AMI.

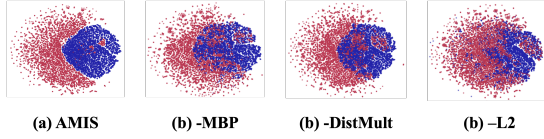


Figure 5: T-SNE visualizations of the representations before the link predictor that are learned by AMI and its three variants on the Alibaba dataset.

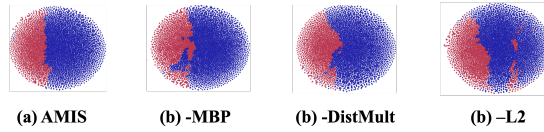


Figure 6: T-SNE visualizations of the representations before the link predictor that are learned by AMI and its three variants on the DBLP dataset.

in link prediction tasks, but also exhibits excellent generalization capabilities across diverse datasets.

5.3 Ablation Study

To further evaluate the effectiveness of our AMI, we conducted an ablation study on three variants of AMI. We adopt two evaluation metrics, i.e., AUC and PR, and conduct experiments on the Alibaba, DBLP, IMDB and Alibaba-s datasets for link prediction tasks. And the experimental results are shown in Figure 4, Figure 5 and Figure 6. Specifically, we generate the following variants.

(i) **w/o MBP (-MBP)**: In this variant, we remove the enhanced interaction adjacency matrix $A_{interact}$. W/o MBP indicates that neighborhood aggregation relies only on the basic behavior pattern matrix $A_{pattern}$, which prevents AMI from capturing the mutual promotion and mutual inhibition effects among multi-interactions in the multi-relational heterogeneous network.

(ii) **w/o DistMult (-DistMult)**: In this variant, we remove the adaptive relation-specific decoder and replaced it with a general dot-product decoder. W/o DistMult indicates that all relation types share the same prediction function, leading to the loss of relation-specific modeling capacity.

(iii) **w/o L2 norm (-L2)**: In this variant, we remove the L2 normalization applied during the training process so that the AMI is optimized without any regularization to prevent overfitting.

Then, we analyze the results of the ablation study from different perspectives, with quantitative analysis presented in Section 5.3.1, and qualitative analysis presented in Section 5.3.2.

5.3.1 Quantitative Analysis. We evaluate the performance of AMI and its three variants in several real-world datasets, in terms of AUC and PR as evaluation metrics, with the results presented in Figure 4. The results indicate a significant decline in model performance for the ablation variants, underscoring the importance of each component. By comprehensively examining the data in Figure 4, we obtain the following conclusions:

(1) **-MBP**. We observe that AMI-MBP exhibits a noticeable decrease in performance compared to the full AMI. In the Alibaba dataset, its performance decreased by approximately 3.9% and 2.1% in AUC and PR. In addition, its performance also decreased by approximately 1.7%, 3.26% and 1.25% in AUC, and decreased by 7.9%, 2.52% and 2.08% in PR on the other datasets. These experimental results demonstrate that the design of the enhanced interaction adjacency matrix $A_{interact}$ plays an essential role in improving the accuracy of link prediction. And these experimental results validate our assumption that multi-relationships actually exhibit characteristics of mutual promotion and mutual inhibition and should not be treated independently.

(2) **-DistMult**: Compared with the full model, this variant shows a drop in performance on all datasets in Figure 4. These experimental results demonstrate that the decoder without $s_{dist}(\cdot)$ is insufficient to capture differences of diverse relations. Based on our hybrid scoring strategy, AMI can provide more precise results for multiple relationships.

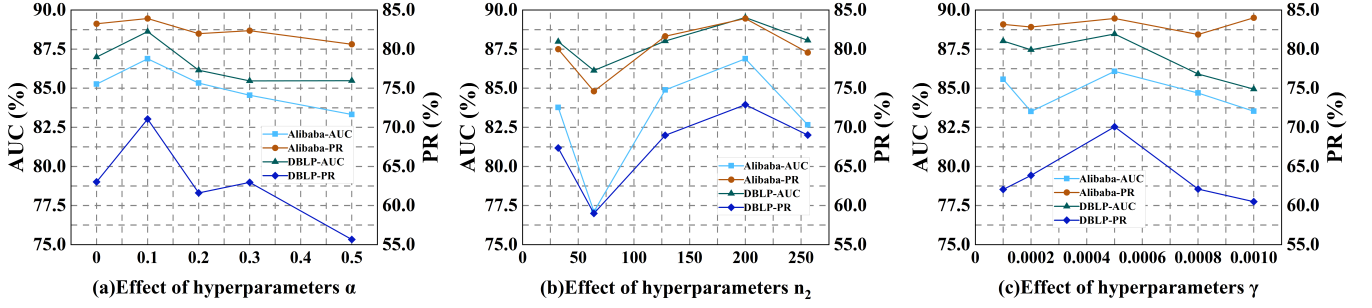


Figure 7: Hyperparameter sensitivity analysis of AMI.

(3) **-L2**: As shown in Figure 4, the -L2 variant plays the most important role of our AMI. The value of AUC significantly drops on the Alibaba and Alibaba-s datasets, which demonstrates that the L2 regularization can not be ignored especially in the multiplex bipartite networks. In addition, L2 regularization can effectively mitigate overfitting, stabilize the training process, and further enhance the generalization ability of the model.

5.3.2 Qualitative Analysis. To qualitatively analyze the effectiveness of AMI, we use t-SNE [35] to visualize the representations before the link predictor that are learned by AMI and its three variants on the Alibaba and DBLP datasets. From Figure 5 and Figure 6, we observe a distinctive boundary between different label clusters in AMI compared to its variants. These results indicate that the representations learned by AMI are more separable and informative. This enhanced discriminative ability suggests that AMI can effectively capture and utilize meaningful features for link prediction tasks.

5.4 Parameter Sensitivity

In this section, we further evaluate the hyperparameter sensitivity of the AMI on the link prediction task. The experiments focus on three hyperparameter: the parameter α that balances the contribution between the general similarity score and the relationship-specific score in the decoder, the latent embedding dimension n_2 , and the parameter γ that controls the importance of L2 regularization in the final loss function $\mathcal{L}_{total}$.

Effect of relation-specific scoring intensity α . As shown in Figure 7(a), the performance of AMI increases when the parameter α increases from 0 to 0.1. And the best performances in the Alibaba and DBLP datasets are when the α is 0.1, indicating that Distmult is efficient for AMI for various types of relations. In addition, when α exceeds 0.2, values of AUC and PR all tend to gradually decrease, demonstrating that capturing the general similarity of nodes in the embedding space is more important than considering relation-specific inductive biases. The experimental results demonstrate that the large relation-specific score may amplify noise and lead to overfitting, while a moderate value of α enables AMI to better capture the different contributions of relation semantics for link predictions.

Effect of embedding dimension n_2 . As shown in Figure 7(b), we observe that various values of embedding dimension n_2 have

different impacts of our AMI. If the embedding dimension n_2 increases from 32 to 200, the performance of our AMI will improve on the four datasets. The best performance of four datasets is achieved when the embedding dimension n_2 reaches 200. For example, when n_2 equals to 200, the values of AUC and PR are 86.88% and 83.91% in the Alibaba dataset; in the DBLP dataset, they are 89.52% and 72.89% respectively. However, if the dimension is further increased, the performance of AMI will drop significantly in all datasets. The experimental results demonstrate that the high value of embedding dimension may introduce noise.

Effect of regularization strength γ . As shown in Figure 7(c), the performance of AMI remains relatively stable when γ ranges from 0 to 0.001. The best performance is obtained when $\gamma = 0.0005$ on all datasets. However, if γ continuously increases, the performance of our AMI will decrease significantly. The reason is that overwhelmingly relying on L2 regularization will impose extremely constraining constraints on the training process, resulting in the loss of significant information. The experimental results imply that the proper hyperparameter γ provides an effective way to balance complexity control and information preservation.

6 Conclusion and Future Work

In this work, we propose an adaptive multi-interaction representation method named AMI, which effectively captures the characteristics of mutual promotion or mutual inhibition for multi-relational web semantic graph. Specifically, AMI mainly consists of three modules: a multi-interaction aggregation module, a global pattern aggregation module, and an adaptive relation-specific decoder module. Experiment results on several real-world multi-relational heterogeneous graph datasets demonstrate the superiority of our AMI in link prediction tasks. In the future, we explore the possibility to empower our method into Large Language Models (LLMs) by converting multiplex heterogeneous graphs into ordered token lists, aiming at representing heterogeneous web semantic graph more effectively. In addition, we will attempt to extend AMI to dynamic multi-relational heterogeneous networks for temporal link prediction tasks.

Acknowledgments

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A Appendix

A.1 Notations and Definitions

The key notations and their descriptions of this paper are listed in Table 2.

Lift Value, also known as the interest factor, is to compute the conditional probability of two events with their prior probability to determine the strength of the association between the events. In this paper, the lift value between two relationships is defined as $Lift(A \rightarrow B) = \frac{P(B|A)}{P(B)} = \frac{P(A \cap B)}{P(A) \times P(B)}$, where $P(\cdot)$ denotes the probability of a relationship occurring, and $P(B|A)$ represents the conditional probability of relationship B occurring given that relationship A has already occurred. When the $Lift(A \rightarrow B)$ is greater than 1, it indicates that relation A and relation B are mutual promotion; when $Lift(A \rightarrow B)$ is less than 1, it indicates that relation A and relation B are mutual inhibition; otherwise, it indicates that relation A and relation B are independent.

Table 2: Notations and Definitions.

Notation	Definition
$\mathcal{G}=(\mathcal{V}, \mathcal{E})$	A multi-relational heterogeneous network (or graph)
n_2	Latent embedding dimension
N	Total types of nodes in $\mathcal{G}$
R	Total types of relationships in $\mathcal{G}$
$\mathbf{v}_s, \mathbf{r}_i, \mathbf{v}_t$	Vector of node v_s , relation r_i and node v_t
$\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3$	Learnable parameter matrices
$\mathbf{X}$	Node feature matrix of $\mathcal{G}$
L	Total number of convolution layers
$\hat{\mathbf{H}}_{\text{final}}$	Final node embedding matrix
$\mathbf{A}_{r_i}$	Basic Behavior Pattern matrix of relation r_i
α_i	Learnable parameter for relation r_i
$\mathbf{A}_{\text{pattern}}$	Basic behavior pattern matrix
$\mathbf{A}_{\text{interact}}$	Enhanced interaction adjacency matrix
$\mathbf{H}_{\text{neigh}}$	final neighbourhood aggregation output
$\mathbf{I}_{\text{cond}}$	Condition interaction matrix
$\mathbf{A}_{\text{GBPM}}$	Global behavior matrix

A.2 Pseudo-Code of AMI

The learning process of AMI is shown in Algorithm 1.

Algorithm 1: The Learning Process of AMI

Input: Multi-relational heterogeneous graph $\mathcal{G}$
Output: Node embedding matrix $\hat{\mathbf{H}}_{\text{final}}$

- 1 Generate basic behavior matrices $\{\mathbf{A}_{p=1,2,\dots,|R|}\}$ from the input graph $\mathcal{G}$.
- 2 Calculate $\mathbf{A}_{\text{pattern}}$ and $\mathbf{A}_{\text{interact}}$.
- 3 Construct $\mathbf{A}_{\text{MBP}} = \mathbf{A}_{\text{pattern}} + \lambda \mathbf{A}_{\text{interact}}$ using Eq. (5).
- 4 **for** $i = 1$ **to** L **do**
- 5 $\mathbf{H}_{\text{neigh}}^{(i)} \leftarrow \mathbf{A}_{\text{MBP}} \cdot \mathbf{H}_{\text{neigh}}^{(i-1)} \cdot \mathbf{W}_3^{(i)}$ using Eq. (6).
- 6 **end for**
- 7 $\mathbf{H}_{\text{neigh}} \leftarrow \frac{1}{L} \sum_{i=1}^L \mathbf{H}_{\text{neigh}}^{(i)}$ using Eq.
- 8 Calculate $\mathbf{A}_{\text{GBPM}}$ using Eq.(10).
- 9 **for** $i = 1$ **to** L **do**
- 10 $\mathbf{H}_{\text{homo}}^{(i)} \leftarrow \mathbf{A}_{\text{GBPM}} \cdot \mathbf{H}_{\text{homo}}^{(i-1)} \cdot \mathbf{W}_{\text{homo}}^{(i)}$
- 11 **end for**
- 12 $\mathbf{H}_{\text{final}} \leftarrow \frac{1}{2} (\mathbf{H}_{\text{neigh}} + \mathbf{H}_{\text{homo}})$ using Eq.(11).
- 13 $\hat{\mathbf{H}}_{\text{final}} \leftarrow \|\mathbf{H}_{\text{final}}\|_{l2}$.
- 14 Calculate the total loss $\mathcal{L}_{\text{total}}$ using Eq.(17).
- 15 Back propagation and update parameters in AMI.
- 16 **Return** $\hat{\mathbf{H}}_{\text{final}}$.
